

Balaji Chidambaram

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Principal Consultant, AI Engineer at Genpact. I take multi-agent platforms from prototype to production for enterprise teams - and make them auditable once they are there.

EXPERIENCE

Genpact - Principal Consultant, AI Engineer

Mar 2026 - Present

Remote, USA

- Lead architect for an enterprise Agentic xP&A platform unifying Financial Planning, Supply Chain, Workforce Planning and Demand Planning across 4 business segments. Ran client discovery, set the multi-agent reference architecture, and shipped a production MVP on Databricks in 1 month with a cross-functional team of 3 engineers and 2 SMEs.
- Architected a Lakebase-backed agent governance layer where finance users curate 400+ KPIs and drivers directly, with versioned lineage - clone, edit, publish, audit. Cut data-readiness prep from over a month to a few days, with every agent output traceable to a governed source.

Genpact - Assistant Manager, AI Engineer

May 2024 - Feb 2026

Remote, USA

- Architected a multi-agent Conversational AI platform using a LangGraph orchestrator over 8 deterministic tools for natural-language analysis, reducing analysis lead time from days to minutes for 100+ enterprise users. A master orchestrator routed requests to domain sub-agents via hierarchical delegation, ReAct, and Supervisor patterns.
- Exposed 20+ reusable analysis skills through MCP servers with least-privilege tool access and prompt-injection guardrails on every tool call, letting domain and agent teams build in parallel.
- Architected a persistence layer for agentic workflows enabling longitudinal, multi-turn analysis across reporting cycles - shifting the product from one-shot Q&A to continuous conversational intelligence.
- Systematized agent quality assurance with an MLflow GenAI evaluation harness using LLM-as-a-judge scorers, converting subjective review into regression-tested SLAs - raising user acceptance from 40% to 60% and preempting ~20 pre-production regressions.
- Engineered a query-lineage governance layer surfacing underlying SQL and source datasets, moving AI-driven analysis from a black box to a fully auditable enterprise solution.

Keck Medicine of USC - Data Engineer / Marketing Data Analyst

Sep 2022 - Dec 2024

Los Angeles, CA

- Engineered a multi-source data warehouse and analytics platform on AWS Redshift, unifying analytics across data silos.
- Built AWS-based ingestion and ETL pipelines (Lambda, EC2, VPC, RDS) that cut reporting prep time by 40%, and designed a Tableau/AWS/Looker analytics platform that raised campaign visibility by 54% and engagement by 32%.

Sayari Labs - Data Engineer Intern

May 2023 - Jul 2023

Washington, D.C.

- Built scalable microservice-style data pipelines (Scrapy, Kubernetes, Airflow) ingesting and transforming financial and corporate data from 10+ sources monthly with high availability.

USC Information Sciences Institute - Research Assistant, Center on Knowledge Graphs

Aug 2022 - May 2023

Los Angeles, CA

- Executed knowledge graph algorithms with the KGTK toolkit, achieving 78% accuracy predicting information flow.
- Developed a chatbot using Named Entity Recognition and Relation Extraction with Transformers, converting GitHub READMEs into knowledge graphs and improving query response accuracy by 40%.

HertzAI - Cloud Architect Intern

Aug 2021 - Nov 2021

Chennai, India

- Automated the CI/CD pipeline with Jenkins across ten cloud services, improving release cycles by 40% and reducing deployment errors by 30%.
- Architected and optimized an API Gateway managing traffic surges and halving response times.

EDUCATION

M.S. Computer Science - University of Southern California

Jan 2022 - Dec 2023

GPA 3.81 / 4.0

B.E. Computer Science and Engineering - Anna University, India

Aug 2017 - Jul 2021

GPA 8.97 / 10

PUBLICATION

P. Shekhar, R. Saha, M. J. B. Dudekula, B. Chidambaram. "Effectiveness of Retrieval Augmented Generation, Contextualized Examples and Prompt Finetuning on Data Enrichment, Cleaning and Master Data Creation"*IEEE ACDSA*, 2026. <https://doi.org/10.1109/ACDSA67686.2026.11467982>

PROJECTS

- **Automated Stock Market Streaming** - Real-time stock market streaming with Kafka, Apache Spark and Cassandra, orchestrated in Airflow. 30% faster processing, 40% lower retrieval time.
- **KG + Fusion Transformer for Multi-Hop QA** - Fine-tuned RoBERTa to improve embedding quality in Dynamically Fused Graph Networks; replaced skip-connections with Compact Bilinear Pooling and Tucker Fusion for faster convergence.
- **Adapting Vision-Language Models to Vision-Only Tasks** - Optimized Vision-Language models for object detection, with Docker and GCP for reproducible training and deployment plus a real-time streaming pipeline.

SKILLS

- **Specialties:** Agentic AI & Multi-Agent Systems (ReAct, Hierarchical Delegation, Supervisor), MCP, Agent Skills, Distributed Systems Design, RAG & Vector Databases (pgvector), Agent Governance & Lineage, LLM Evaluation & Observability, Knowledge Graphs, ML Forecasting
- **Languages:** Python (PySpark), SQL, C++, C, Java, React
- **AI / ML:** LangGraph, MCP, Google ADK, TensorFlow, PyTorch, MLflow, TimeGPT, KGTK, Transformers
- **Platforms & Operations:** AWS (Lambda, EC2, VPC, RDS, Redshift), Databricks, Google Cloud (Vertex AI, Gemini, Document AI), Azure, Kubernetes, Docker, Airflow, Jenkins, Kafka, Cassandra, CI/CD
- **Certifications:** Google Cloud Professional Machine Learning Engineer, Databricks Generative AI Associate, Google Cloud Associate Cloud Engineer